TEVIX — Feasibility & Technical Report

Smart biodegradable food packaging: bioplastic substrate + embedded spoilage/contamination sensors + AI/ML shelf-life prediction

Prepared for Team Sentinel · Round 2 evaluation runway (brief due 30 June 2026) · Primary market: India
Evidence base: two adversarially fact-checked web-research passes (48 claims verified, 2 refuted). Every figure is tagged by source quality. Nothing is invented — unfound numbers are marked as gaps.

Confidence / source tags used throughout: PRIMARY peer-reviewed paper or government/legal text SECONDARY industry / news / market-research aggregator HIGH MED LOW verification confidence

0 · Executive verdict (read first)

The science of every TEVIX layer is real and lab-proven. None of it is yet a solved, cheap, mass-market embedded product. The binding constraint is not physics — it is cost, long-term stability, food-matrix interference, and (in India) the total absence of regulation.

Recommended feasible V1: ONE chemiresistive/colorimetric gas-freshness sensor (TVB-N / biogenic amines / CO_2) + battery-less NFC readout (phone-powered) + a thin AI shelf-life model, on ONE high-value vertical (meat/seafood or pharma cold-chain). The "all-food + embedded pathogen detection + biodegradable + pennies-per-unit" vision is **not feasible today** and should be presented as a staged R&D roadmap.

Feasibility axis	Verdict	The number that decides it
Physical / sensor	FEASIBLE (lab)	Pathogen biosensors: single-digit CFU/mL in 20–80 min; MEMS readout proven across 4+ decades at 0.98 mW
Power	FEASIBLE but constrained	Battery-less NFC tag <1 mW; smartphone supplies several mW — works only at ≤1–7 cm
Cost	BORDERLINE → unsolved	Smart materials = **50–100%** of packaging cost vs a **<10%** (ideally ¢1/unit) viability target
Regulatory	VACUUM in India	FSSAI 2018 has **zero** provisions for smart/active/intelligent packaging; draft amendment only entered comment period Feb 2026

1 · Problem statement & its significance (India)

Conventional food packaging is **passive** — it cannot report, in real time, whether the food inside is spoiling or contaminated. The economic consequence in India is large and **government-quantified** (not estimated by us):

Metric	Figure	Source PRIMARY
Total annual post-harvest loss, India	**₹92,651 crore ≈ USD 17.1 billion** (production year 2012–13, 46 commodities)	ICAR-CIPHET 2015 (Jha et al.), via ICRIER HIGH
Share of loss occurring at farm/early stage	~77–87% of total loss	ICAR-CIPHET 2015 HIGH
India total agricultural exports	**USD 53.1 billion** (FY2022–23); APEDA channel USD 26.7 bn	PIB / Ministry of Commerce 2024 HIGH

Per-commodity post-harvest loss — where spoilage sensing matters most

NABCONS 2022 (latest MoFPI-commissioned pan-India study) vs ICAR-CIPHET 2015. HIGH confidence, government primary source.

Commodity	NABCONS 2022 loss %	CIPHET 2015 loss %	Relevance to TEVIX
Fruits	6.0–15.05%	6.7–15.88%	★★★ High value, fast spoilage, ethylene markers
Vegetables	4.87–11.61%	4.58–12.44%	★★★ High volume loss
Marine fisheries	8.76%	10.52%	★★★ Clear TVB-N / amine markers, safety-critical, export value
Poultry	5.63%	6.74%	★★ Safety-critical
Cereals	3.89–5.92%	4.65–5.99%	★ Low value, price-sensitive
Meat	2.34%	2.71%	★★★ Strong gas markers, high margin
Milk	0.87%	0.92%	★ Low loss %, but matrix interference is hard

> **Honest gap:** The exact *value/rate of India export rejections* due to quality/contamination (EU RASFF, US FDA refusals) was searched but **not found in a verifiable source**. Do not put a rejection-rate number on a slide until it is sourced. Total export value ($53.1B) is solid; the rejection slice is not.

2 · Scope of the problem — which products, which beachhead

Do **not** try to solve all food. The unit economics (Section 5) only tolerate a sensor cost on **high-value, high-spoilage, safety-critical** categories. Recommended priority:

Rank	Vertical	Why it wins / loses
1	**Meat / seafood**	Highest spoilage + clearest gas markers (TVB-N, H_2S, biogenic amines) + safety-critical → tolerates a higher sensor cost. Marine loss 8.76%.
2	**Pharma / cold-chain**	High margin; monitoring is already paid for; best unit economics. Adjacent but most fundable.
3	Fresh fruits / vegetables	Largest loss volume (up to 15%) **but** price-sensitive → cost gap is brutal.
4	Dairy	High value but documented **food-matrix interference** makes sensing hard.
5	RTE / bakery / baby food	Safety-critical niches, smaller volume.

3 · Market size

> **Source-quality warning — read before quoting these.** Unlike Sections 1–2, the market-size figures below come **only from commercial market-research aggregators** (Precedence, GMInsights, MarketsandMarkets, PHDCCI). They are paywalled, methodologically opaque, and **did not survive our independent fact-check**. Treat them as *directional*. They are fine for a "TAM is large" slide; do **not** present them as verified fact.

Market	Base year	Forecast	CAGR	Source SECONDARY
India food processing industry	USD 307 bn (2023)	USD 700 bn by 2030	~12.5%	PHDCCI via IBEF MED
Global active & intelligent packaging	USD 14.9 bn (2025)	USD 41.4 bn (2035)	11%	GMInsights LOW
Global smart packaging	USD 32.2 bn (2025)	USD 55.6 bn (2034)	6.24%	Precedence Research LOW
Global biosensors	USD 34.5 bn (2025)	USD 54.4 bn (2030)	9.5%	MarketsandMarkets LOW

The most defensible market framing is **not** these TAM numbers — it is the government-verified ₹92,651 cr / $17.1B annual loss (Section 1), which is the real money your product addresses.

4 · Technical feasibility

4.1 Sensing layer — feasible in lab, research-stage for embedding

Sensor class	Detects	Readiness	Verified performance
Chemiresistive / electrochemical gas	TVB-N, biogenic amines, H_2S, CO_2, ethylene, pH	Lab-mature; **most embeddable**	Best fit for flexible, low-power, printable packaging PRIMARY HIGH
Colorimetric freshness indicators	Same markers via colour change	A few commercial; mostly lab	Cheap, no electronics — but yes/no only, no data, no prediction
Pathogen biosensors (E. coli, Salmonella, Listeria)	Specific pathogens	Lab assays / benchtop only	**Single-digit CFU/mL** (Salmonella ~3, E. coli O157:H7 2–4, Listeria 7.1) in **20–80 min** HIGH
RFID / NFC sensor tags	ID, temp, tamper, scan-time reads	Commercial	Tracking/auth; limited true freshness sensing

Not solved: Embedding a real pathogen biosensor into a cheap wrapper does not exist commercially. Only 1 of 77 reviewed studies tested naturally contaminated food; dairy and meat matrices are explicitly cited as interference-prone. Treat pathogen detection as roadmap, not V1.

4.2 MEMS resistive sensing + analog readout — the evaluator-facing core

Your Round-2 evaluator specializes in **analog/mixed-signal VLSI, resistive MEMS sensors, and sensor-interface signal conditioning.** The literature directly proves the readout chain is feasible — lead with this:

- A tiny resistive change in a MEMS gas sensor can be read across **~100 Ω to 10 MΩ (4+ decades)** at **sub-milliwatt power (~0.98 mW @ 3.3 V).** PRIMARY HIGH
- Proven architecture blocks: **Wheatstone-bridge / differential front-end** (ratiometric read) → **chopper-stabilized amplifier** (suppresses DC offset + dominant 1/f flicker noise, the key problem at high sensor resistance) → **instrumentation amp** (selectable gain 6–42 dB) → **mismatch-correction trim DAC** → **reconfigurable SAR / ΔΣ / single-slope ADC** (resolution-vs-power tradeoff). Fabricated nodes: 0.18 µm and 110 nm CMOS.

The mechanism paragraph to rehearse: "The gas-sensitive film's resistance shifts a few percent over a wide baseline. We read it ratiometrically with a Wheatstone bridge, chopper-stabilize the front-end to push 1/f noise and offset out of band, amplify with an instrumentation amp, trim mismatch with a correction DAC, then digitize with a reconfigurable SAR/ΔΣ ADC — keeping the whole chain sub-milliwatt so it runs off harvested NFC power."

Two honesty caveats — state them, don't hide: (1) one key wide-range readout paper is **post-layout simulation, not taped-out silicon** — call it "simulated." (2) A "160 dB / 0.1% over 5 decades resistance-to-time converter" claim **failed our fact-check (refuted 1–2)** — do **not** cite it.

4.3 AI / ML shelf-life layer — feasible & demonstrated

- TinyML on an **Arduino Nano 33 BLE Sense** + multichannel gas-sensor array reached **91.9% accuracy (AUC 0.98)** for shelf-life estimation in cold and ambient storage. PRIMARY HIGH
- Framing discipline: present as "a calibrated regression on sensor + temperature time-series," **not** "AI magic." A hardware evaluator respects rigor over buzzwords.

4.4 Power — feasible but distance-constrained

Metric	Verified figure	Source PRIMARY
NFC energy harvest from phone	~5 mA @ 2–3 V (advanced ICs up to 10 mA / ~15 mW)	Sensors 2018 survey HIGH
Battery-free spoilage tag power draw	**<1 mW** (vs >10 mW for ordinary gas-sensor approaches)	Nature Sci. Reports 2019 HIGH
Demonstrated NFC sensor tag current	Draws ~3 mA < 5 mA harvestable → **no battery needed**	Sensors 2019 (Lazaro) HIGH
Practical read range	~1–7 cm, falls to ~0 beyond; phone-model dependent	peer-reviewed HIGH

Conclusion: phone-powered battery-less NFC is the only economically plausible power route for consumer packs, and it is demonstrated in the lab — but only for ultra-low-power (<1 mW) designs read at close range. Printed-battery / energy-harvesting cost trajectory for passive packs was **not** found (gap).

5 · Cost feasibility & accessibility — the make-or-break section

Component / threshold	Verified cost	Source
Viability threshold (max smart-feature cost)	Printed sensor must add **≤ ~US ¢1** per unit; broader rule: smart feature <10% of packaging cost	ACS Sensors 2022 (Barandun, Imperial) PRIMARY MED (scoped to poultry)
Current reality of smart materials	Run **50–100%** of total packaging cost (lab/prototype)	Foods/MDPI review PRIMARY HIGH
Passive UHF RFID tag	USD **0.09–0.18** / unit at 100k volume (–17–33% from 10k→100k)	Future Foods 2022 PRIMARY HIGH
Full battery-less NFC sensor tag (BOM)	**< USD 5** at high volume (colour sensor $1.5 + NFC IC $0.3 + MCU $1.0 + case)	Sensors 2019 PRIMARY HIGH

The central feasibility gap, stated honestly: a complete sensor tag today is **~$5 BOM**, but mass-market food packaging needs the smart feature at **~¢1 (≈ $0.01)** — a **~500× gap**. This is why V1 must target high-value verticals where a $1–5 tag is tolerable, not commodity produce. Closing to ¢1 depends on printed/flexible-electronics scale economics, which is plausible long-term but unproven.

Honest gap: standalone per-unit costs for colorimetric labels, chemiresistive gas sensors, MEMS gas sensors, pathogen biosensors-per-test, and TTI labels were searched but **not found in verifiable sources**. Do not invent them.

6 · Regulatory feasibility

Jurisdiction	What applies	Key numbers / status PRIMARY legal text
EU	Reg (EC) 1935/2004 (framework) · Reg (EC) 450/2009 (active & intelligent) · Reg (EU) 10/2011 (plastics)	Intelligent materials = those that **monitor food condition** (TEVIX fits here) and **must not release constituents into food**; only "Community list" substances allowed. Overall Migration Limit **60 mg/kg (10 mg/dm²)**. HIGH
India (FSSAI)	FSS (Packaging) Regulations 2018, Version-V (02 Apr 2025)	OML **60 mg/kg** per IS 9845; SMLs e.g. Ba 1.0, Co 0.05, Cu 5.0, Zn 25.0, DEHP 1.5 mg/kg; rPET permitted (amendment **28 Mar 2025**). HIGH
US (FDA)	Food-Contact Substance (FCS) / FCN notification regime	Intelligent-packaging-specific FCN requirements requested but **not verified here** (gap).

THE most important regulatory finding for TEVIX: India's in-force FSSAI Packaging Regulations 2018 contain **ZERO** provisions for active, intelligent, or smart packaging, embedded sensors, indicators, or biodegradable food-contact materials (full-text extraction: zero hits for "active/intelligent/smart/sensor/indicator/biodegrad"). A **draft 2026 amendment** proposing to add "active and intelligent materials" only entered its 60-day comment period (26 Feb 2026) and is **not in force**. **TEVIX would launch into a regulatory vacuum in India** — simultaneously an opportunity (first-mover, shape the standard) and a risk (no approval pathway; sensor must still meet the 60 mg/kg migration limit and not contaminate food).

7 · Competitive landscape & differentiation

Existing players fall into three buckets — **none** covers all of TEVIX. This gap is your story.

Bucket	Players	What they're missing
Sense, but don't predict	RipeSense, SensorQ, FreshTag, Senoptica (O_2 colour), Mimica	No data capture, no AI, often yes/no only
Benchtop lab instruments	3M Molecular Detection System, bioMérieux VIDAS	Not embeddable in packaging
Connected / predictive but not cheap-embedded-biodegradable	Evigence (raised USD 18M), Blakbear (gas label + "Freshness API"), Avery Dennison RFID	Cost; not compostable; not all-in-one

TEVIX whitespace: cheap + embedded + predictive + biodegradable, all at once — which is also exactly why it's hard (cost physics, Section 5). **Closest direct competitor: Blakbear.** Differentiate sharper than "Blakbear but biodegradable."

Category trap to avoid: Industrial-automation sensor vendors (Leuze, Baumer, SICK, Pepperl+Fuchs, Holykell) shared in earlier team research are **line-inspection sensors on factory machines** (detect "is a carton present?"), **not** in-package food sensors. Citing them signals a category error to a hardware evaluator. Exclude them.

8 · Is it implementable in real life? — final synthesis

Yes — as a scoped V1, not as the full vision.
Buildable today: a phone-read, battery-less NFC tag carrying a chemiresistive/colorimetric gas-freshness sensor with a calibrated shelf-life model, on meat/seafood or pharma cold-chain, at a $1–5 tag cost that those verticals can absorb. Every piece of that is lab-proven and primary-sourced.

Not buildable today: embedded pathogen detection in a wrapper; ¢1-per-unit cost on commodity produce; sensors fully integrated into certified compostable bioplastic; an approved Indian regulatory pathway. These are the roadmap — and naming them honestly is what makes the team credible to a hardware evaluator.

9 · Honest evidence gaps (your research to-do before 30 June)

- Verified market-size numbers (current figures are aggregator-grade only).
- India food **export-rejection** value/rate (EU RASFF, US FDA refusals).
- Standalone per-unit costs: colorimetric label, chemiresistive gas sensor, MEMS gas sensor, pathogen biosensor/test, TTI label.
- Printed-battery / flexible-electronics cost trajectory.
- US FDA FCN requirements specific to intelligent packaging.
- Confirm in-force status of India's draft 2026 smart-packaging amendment.

Full source bibliography with briefs and outcomes follows in the companion file `TEVIX_Research_Sources.pdf` (and the section below).